Predicting Customer Term Deposit Subscription Using Supervised Machine Learning

Final Project Report

Dataset: UCI Bank Marketing Dataset

Modelling focus: Supervised binary classification with imbalance-aware evaluation

This report documents the data preparation, exploratory analysis, model development, evaluation, threshold analysis and business interpretation carried out in the executed project notebook.

1. Executive Summary

This project develops a supervised machine learning approach for predicting whether a bank customer will subscribe to a term deposit. The analysis uses the UCI Bank Marketing dataset, which contains 45,211 customer records and 17 columns, including the binary target variable *y*. The modelling dataset uses 15 predictors after excluding *duration*, because call duration is only known during or after the interaction and can introduce target leakage in a realistic pre-contact prediction setting.

The target is strongly imbalanced: 39,922 customers (88.3%) did not subscribe, while 5,289 (11.7%) did. Consequently, accuracy alone would give an incomplete view of model performance. Logistic Regression, Decision Tree and Random Forest models were evaluated under three approaches: the standard class distribution, class weighting and SMOTENC-based oversampling.

The strongest default-threshold model by F1 score was the balanced Random Forest, with an F1 score of 0.448 at the model's standard operating threshold. A separate threshold analysis showed that, among the tested thresholds, 0.40 produced the highest F1 score: 0.468, with precision of 0.419 and recall of 0.528. This operating point provides a more balanced trade-off between identifying likely subscribers and limiting false positives.

2. Business Problem and Objectives

Business problem. A bank needs to identify customers who are more likely to subscribe to a term deposit so that marketing activity can be targeted more effectively. Because positive subscriptions are relatively uncommon, a useful model should identify a meaningful share of potential subscribers without generating an impractically large number of false positives.

Project objective. Develop and compare supervised classification models, assess their predictive performance using imbalance-aware metrics, select a suitable model for the business context, and interpret the variables that contribute most strongly to the model's predictions.

Analytical questions. (1) How imbalanced is the subscription target? (2) Which modelling approach provides the strongest minority-class performance? (3) Does class weighting or SMOTENC improve the treatment of the minority class? (4) Which predictors contribute most to the selected model? (5) Can changing the probability threshold improve the balance between precision and recall?

3. Dataset and Feature Selection

The UCI Bank Marketing dataset contains customer demographic, financial and campaign-related variables. The target *y* indicates whether the customer subscribed to a term deposit (*yes*/*no*). The analysis used the semicolon-separated *bank.csv* file in the project.

Group	Variables
Numerical	age, balance, day, campaign, pdays, previous
Categorical	job, marital, education, default, housing, loan, contact, month, poutcome
Target	y — term deposit subscription (yes/no)

Leakage control. The variable *duration* was excluded from modelling. Although it is predictive, it represents the length of the current contact and is only available after the interaction has taken place. Including it would make the model less appropriate for a genuine pre-call targeting use case.

4. Data Quality Assessment

Check	Finding	Decision
Standard missing values	No standard NaN values detected	No row deletion or missing-value replacement was required for the raw data

Check	Finding	Decision
Duplicate rows	0 duplicates	No duplicate removal required
Categorical 'unknown'	Present in several categorical fields	Retained as a meaningful categorical state rather than deleting large numbe
pdays = -1	Used to indicate no previous contact	Retained as a campaign-history indicator

5. Exploratory Data Analysis

Exploratory analysis was used to understand the target imbalance and inspect patterns in customer and campaign variables before model fitting. The figures below are the project's generated outputs.

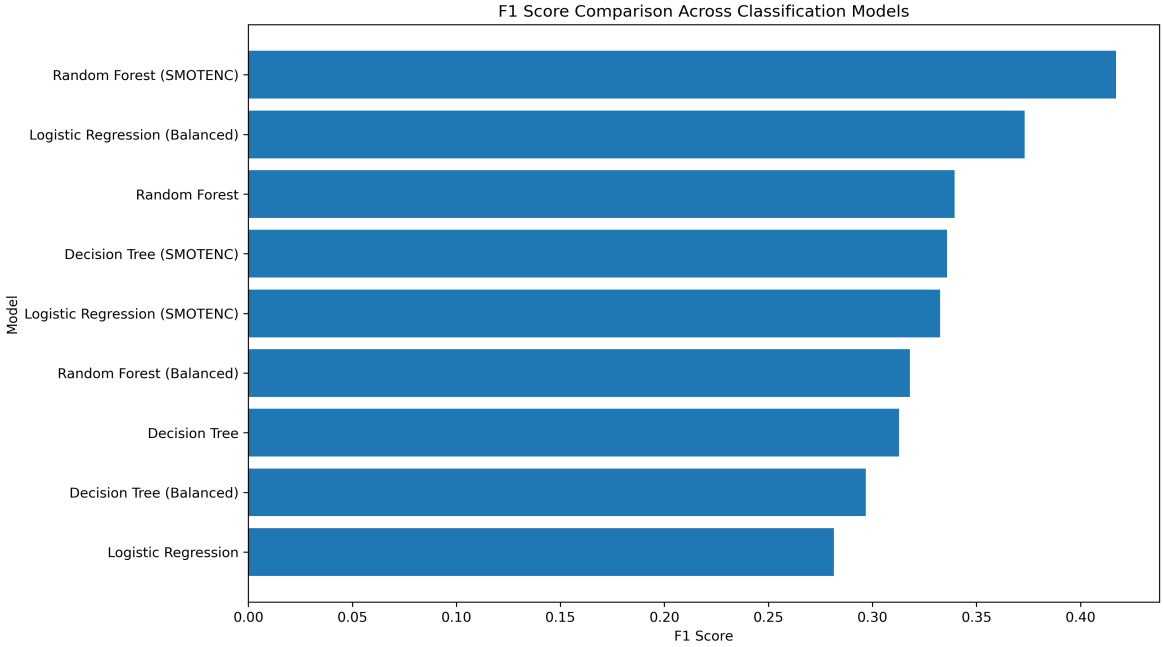


Figure 1. F1-score comparison across the evaluated model configurations.

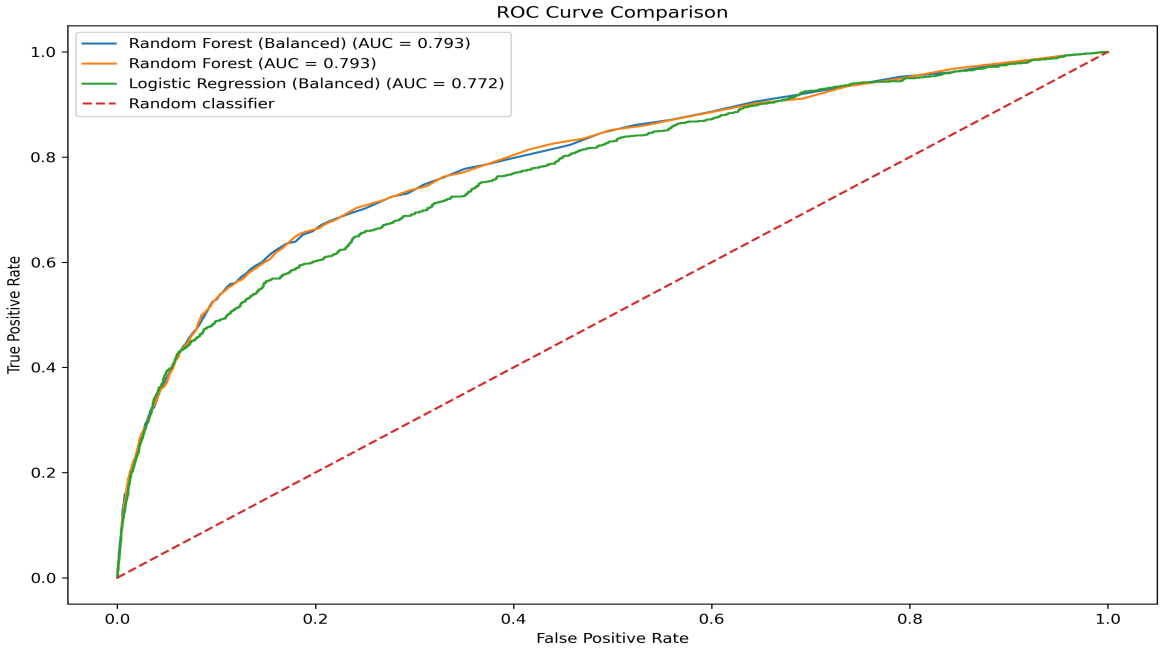


Figure 2. ROC curves for the strongest selected candidate models.

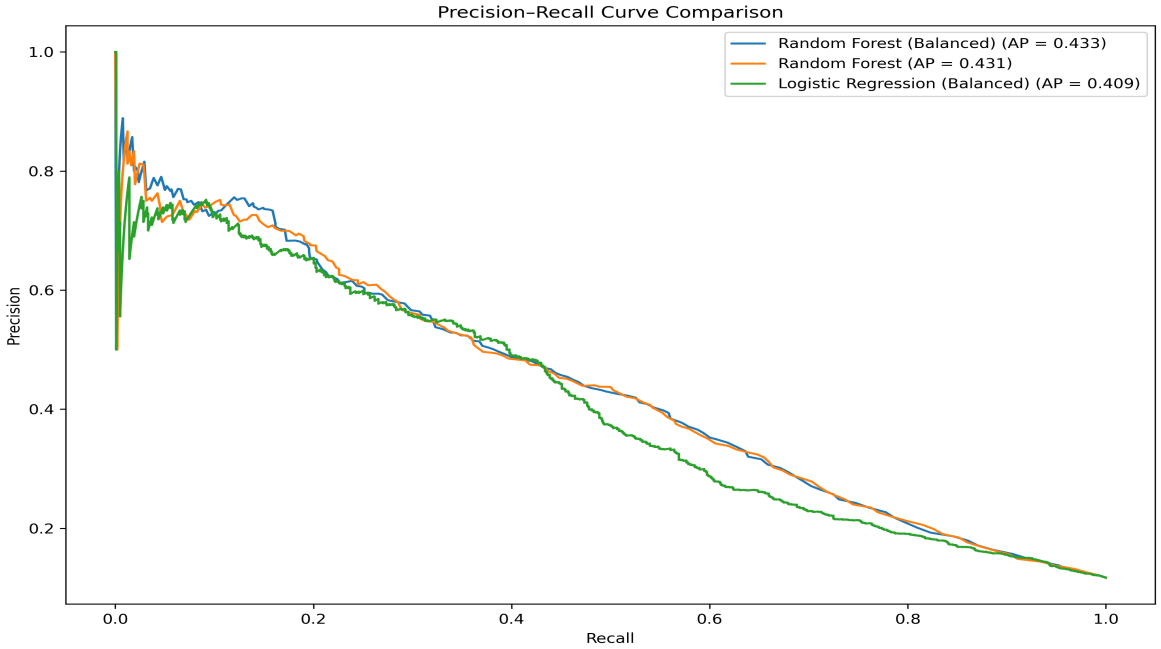


Figure 3. Precision-recall curves for the strongest selected candidate models.

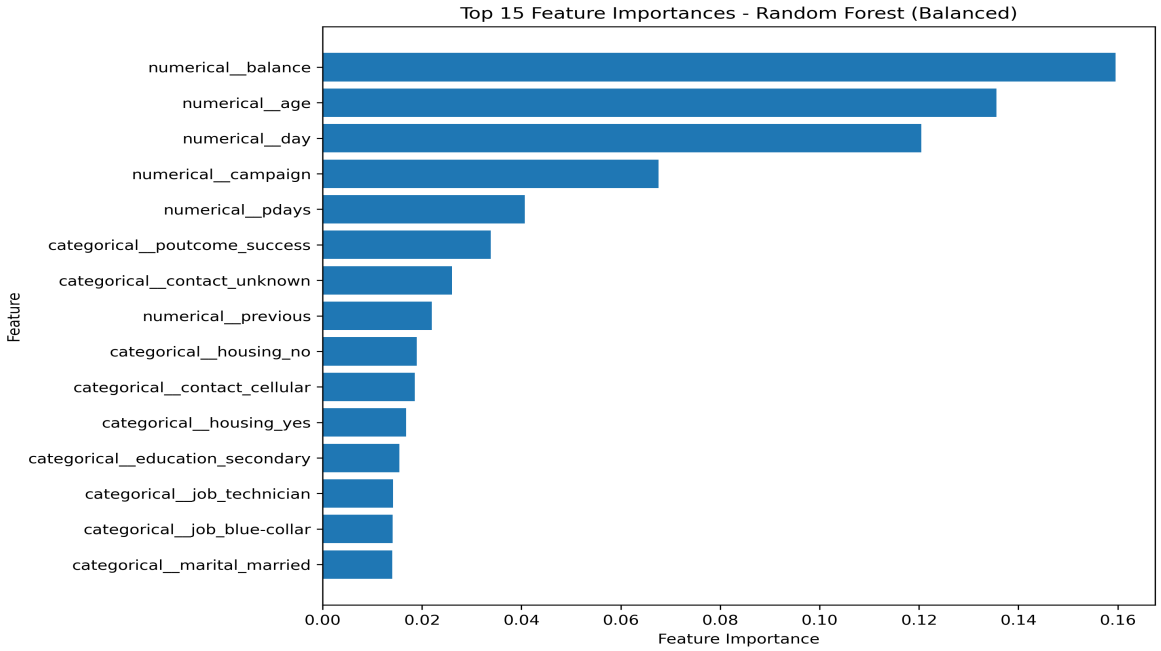


Figure 4. Top feature importances from the balanced Random Forest.

Target imbalance. The positive class represents only 11.7% of observations. A majority-class prediction would therefore achieve approximately 88.3% accuracy without identifying any subscribers. This establishes why recall, precision, F1 and ROC-AUC are more informative than accuracy alone.

Interpretation of exploratory patterns. Campaign history, contact characteristics and customer-level variables contain useful predictive information, but these relationships should be interpreted as associations used for prediction rather than evidence of causation.

6. Preprocessing and Feature Engineering

The modelling workflow separates numerical and categorical processing. Numerical variables are median-imputed and standardised. Categorical variables are imputed using the most frequent category and one-hot encoded with unseen-category handling. This preprocessing is implemented through scikit-learn pipelines so that transformations are fitted using training data and consistently applied to test data.

The target was encoded as 0 for *no* and 1 for *yes*. A stratified 80/20 train-test split was used so that the minority-class proportion was preserved in both partitions.

For imbalance experiments, three strategies were compared. The baseline models retained the original distribution. The weighted models used *class_weight='balanced'*. SMOTENC models used synthetic minority oversampling designed to accommodate mixed numerical and categorical data.

7. Model Development

Three different supervised learning algorithms were evaluated: Logistic Regression, Decision Tree and Random Forest. Each algorithm was tested using the standard training distribution, class weighting and SMOTENC. This produced nine evaluated model configurations and allowed both algorithm choice and imbalance treatment to be assessed.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.893	0.663	0.179	0.281	0.772
Decision Tree	0.831	0.298	0.329	0.313	0.613
Random Forest	0.894	0.622	0.233	0.340	0.793
Logistic Regression (Balanced)	0.755	0.266	0.624	0.373	0.772
Decision Tree (Balanced)	0.840	0.308	0.295	0.301	0.604
Random Forest (Balanced)	0.881	0.488	0.413	0.448	0.791
Logistic Regression (SMOTENC)	0.726	0.232	0.582	0.332	0.738
Decision Tree (SMOTENC)	0.776	0.255	0.474	0.331	0.645
Random Forest (SMOTENC)	0.836	0.355	0.491	0.412	0.779

8. Model Comparison and Selection

The results show a clear trade-off created by class imbalance. The unweighted Logistic Regression model has high accuracy and precision but detects relatively few subscribers. Class weighting substantially increases minority recall, especially for Logistic Regression, but at the cost of lower precision and accuracy.

The balanced Random Forest provides the strongest overall F1 score among the nine configurations (0.448) while maintaining a substantially better recall than the ordinary Random Forest. SMOTENC also improves minority recall for the Random Forest, but its F1 score (0.412) is lower than the balanced Random Forest.

The ordinary Random Forest has the highest ROC-AUC (0.793), while the balanced Random Forest has a very similar ROC-AUC (0.791). Because the project places emphasis on identifying the minority class and balancing precision with recall, the balanced Random Forest was selected rather than choosing solely on ROC-AUC or accuracy.

9. Threshold Analysis

The selected Random Forest produces probabilities rather than only class labels, allowing the operating threshold to be adjusted. The project tested thresholds from 0.30 to 0.60. The best F1 score among the tested thresholds occurred at **0.40**. At this threshold, precision is 41.9%, recall is 52.8%, and F1 is 46.8%.

Threshold	Precision	Recall	F1
0.30	0.337	0.619	0.437
0.35	0.377	0.567	0.453
0.40	0.419	0.528	0.468
0.45	0.457	0.467	0.462

Threshold	Precision	Recall	F1
0.50	0.486	0.417	0.449
0.55	0.504	0.357	0.418
0.60	0.538	0.305	0.390

The 0.40 threshold should be viewed as an operating choice for this project rather than a universal optimum. In a production setting, the threshold should be selected using the relative cost of missed subscribers and unnecessary marketing contacts, and validated on future or temporally separated data.

A small technical detail was also verified during evaluation: 13 test observations received an exact predicted probability of 0.50. This explains the small difference between scikit-learn's class prediction at its default threshold and a manual rule using probability >= 0.50. The discrepancy is a tie-breaking detail at exactly 0.50 and does not indicate a model inconsistency.

10. Feature Importance and Interpretation

The balanced Random Forest identifies several variables as important contributors to predictive performance. The strongest individual features in the fitted model are *balance*, *age*, *day*, *campaign* and *pdays*, followed by campaign-history and contact-related categorical indicators such as *poutcome_success* and *contact_unknown*.

Feature importance indicates how useful a variable is to the fitted predictive model; it does not establish that the variable causes subscription behaviour. The results are therefore best used to guide targeting and further investigation rather than to infer causal relationships.

11. Business and Stakeholder Implications

Prioritise likely subscribers. The model can provide a ranked or probability-based customer list rather than treating every customer as equally likely to subscribe.

Use thresholding to match campaign capacity. A lower threshold increases recall and expands the pool of customers contacted, while a higher threshold produces a smaller, more selective list. The 0.40 threshold provides a useful compromise under the project's F1-based objective.

Monitor campaign efficiency. Because false positives represent marketing effort spent on customers who may not subscribe, precision remains important. Conversely, false negatives represent potential subscribers who are not targeted, making recall important as well.

Support rather than replace business judgement. The model should be treated as a decision-support tool. Campaign managers can combine model scores with budget, contact capacity, customer-experience rules and current campaign strategy.

12. Limitations

First, the evaluation uses a single stratified train-test split, so the reported test metrics represent one sample of model performance. Cross-validation and repeated evaluation would provide a more robust estimate.

Second, the dataset reflects historical campaign behaviour. Customer behaviour and campaign conditions can change over time, so future performance may differ from the reported results.

Third, the threshold was selected using the test-set predictions for this project exercise. For a production workflow, threshold selection should be performed using a validation set or cross-validation and then evaluated once on a final hold-out test set.

Fourth, Random Forest feature importance is useful for ranking predictive contribution but does not provide causal explanations. More detailed interpretability methods could be used when individual customer decisions need to be explained.

13. Future Improvements

Future work could include cross-validation, systematic hyperparameter tuning, probability calibration, a time-based validation design, cost-sensitive threshold optimisation and more detailed interpretability analysis. Additional business information could also be incorporated if it is available before the marketing contact and does not introduce leakage.

14. Conclusion

The project demonstrates a complete supervised classification workflow for an imbalanced customer-marketing problem. The analysis compared three algorithms under three imbalance strategies and showed that model evaluation changes substantially when minority-class performance is considered.

The balanced Random Forest was selected as the strongest overall configuration based on F1 and its practical balance between precision and recall. Threshold analysis further improved the operating trade-off, with 0.40 producing the highest F1 among the tested thresholds. The resulting workflow provides a defensible basis for prioritising customers for term-deposit marketing while making clear that future validation and business-cost analysis would be required before production use.

15. References

Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems, 62*, 22–31.

UCI Machine Learning Repository. Bank Marketing dataset. Dataset used as the source for the project analysis.

Scikit-learn documentation. Classification, preprocessing, model evaluation and pipeline utilities used in the implementation.

Imbalanced-learn documentation. SMOTENC implementation used for the mixed-type minority oversampling experiment.